MIL-C-11390C(GL)

21 February 1966

SUPERSEDING
MIL-C-11390B(QMC)
4 May 1962

MILITARY SPECIFICATION

CLOTH, LAMINATED: COTTON, BALLOON, 3 PLY, AIR RETAINING, CHLOROPRENE

1. SCOPE

* 1.1 This specification covers a laminated coated fabric containing 3 plies of chloroprene coated cotton balloon cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CCC-T-191	- Textile Test Methods.
PPP-P-51	- Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cotton, Silks and Synthetics).

MILITARY

MIL-C-12318	- Cloth, Cotton, Balloon.

STANDARDS

FEDERAL

FED. STD. NO. 406	- Plastics, Methods of Testing.
FED. STD. NO. 601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications, required by suppliers in connection with specific procurement functions, shall be obtained from the procuring activity or as directed by the contracting officer.)

FSC 83057

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3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a 3-yard sample of the coated fabric shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of the responsibility for compliance with all provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items. No change in basic formulation or processing technique shall be made after approval of the preproduction sample without submittal and approval of a new sample.

3.2 Materials.-

3.2.1 Base cloth.- The base cloth shall be balloon cotton cloth conforming to type HH of MIL-C-12318, except that the copper content shall not exceed 0.003 percent and the manganese content shall not exceed 0.0015 percent.

* 3.2.2 Coating compound.- The coating compound shall contain chloroprene rubber, formulated in accordance with established principles to meet the requirements of this specification. No natural rubber shall be used. Up to 20% of other elastomers (such as SBR) may be added to facilitate processing. The use of reclaimed rubber is prohibited.

3.2.3 Dusting powder.- The dusting powder for application to the coated fabric shall be whiting, talc or other finely divided mineral material which does not support mildew growth.

3.3 Coated cloth.- The base cloth specified in 3.2.1 shall be coated on both sides with the coating compound specified in 3.2.2 and shall be laminated to conform to the construction and weight requirements in table I. Sections of biased balloon cloth shall overlap 3/4 inch. The laminated coated fabric shall be fully vulcanized. The laminated coated fabric shall be dusted on both sides with the dusting powder specified in paragraph 3.2.3.

TABLE I - Construction and weight of the laminated coated fabric

Construction	Ounces per sq. yd.
Outside proofing, chloroprene, black	2.5
Cotton balloon cloth, HH, straight	2.0
Gas film, chloroprene	3.5
Cotton balloon cloth, HH, biased	2.0
Gas film, chloroprene	3.5
Cotton balloon cloth, HH, straight	2.0
Inside proofing, chloroprene	3.0
Total weight	18.5
	± 1.0 ounce

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3.4 Performance requirements.- The laminated coated fabric, when tested as specified in table IV, shall conform to the requirements specified in table II.

TABLE II - Performance requirements of laminated coated fabric

Breaking strength, pounds, minimum	
Initial	
Warp (length)	80
Filling (width)	75
After accelerated weathering	
Warp (length)	90% of initial breaking strength
Filling (width)	90% of initial breaking strength
Ply adhesion, lbs. per in. of width, minimum	
Warp (length)	4.5
Filling (width)	4.5
Permeability, liters per sq. meter per 24 hours, maximum	6.0
Accelerated weathering	1/

1/ The laminated coated fabric shall not become stiff and brittle or soft and tacky or show other signs of improper vulcanization.

3.5 Dimensions and put-up.-

3.5.1 Width.- The minimum width of the laminated coated fabric shall be 39 inches unless otherwise specified (see 6.2).

3.5.2 Length and put-up.- The laminated coated fabric shall be put up in rolls as specified in PPP-P-51. The length of rolls shall be not less than 80 or more than 100 yards. The maximum number of splices permitted in any roll shall be 3. Splices in individual plies, except in the center (bias) ply, shall be counted as splices and considered the same as full splices through all 3 plies. Splices occurring in the bias ply only, shall not be counted as splices.

3.6 Marking.- The coated fabric shall be marked in accordance with any one of the methods specified in (a), (b) or (c) below.

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Marking shall be along one edge of the outside proofing and shall extend the full length of the coated fabric. The marking material shall be compatible with the chloroprene compound.

(a) Green thread shall be plainly visible and continuous on the outside proofing within 1 inch from the edge. The thread shall be incorporated in the outside proofing before the curing operation.

(b) A painted or stamped green strip, 1/4 to 1/2 inch wide, shall be not more than 3/4 inch from the edge.

(c) A painted or stamped series of green dots or dashes shall be in a line not more than 3/4 inch from the edge. Diameter of the dots or width of the dashes shall be 1/4 to 1/2 inch. Distance between dots or dashes shall be not more than 3 feet.

3.7 Workmanship.- The finished, laminated coated fabric shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- The preproduction sample submitted in accordance with 3.1 shall be visually inspected for appearance. The sample shall be inspected for all physical and chemical properties in accordance with 4.3.3 to the extent applicable.

4.3 Inspection.- Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated.

* 4.3.1 Components and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, the supplier shall submit a certificate of compliance for the following requirements:

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	Requirement paragraph
Copper content less than 0.003%	3.2.1
Manganese content less than 0.0015%	3.2.1
Coating compound is composed of chloroprene and contains no other synthetic rubber, no natural rubber and no reclaimed rubber	3.2.2
Dusting powder	3.2.3
Weight and orientation of each ply	3.3
Construction of laminated fabric	3.3

4.3.2 In-process inspection.- Inspection shall be made at any point during any phase of the processing to assure that construction of the coated fabric is as specified (see 3.3). The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance.

4.3.3 Examination of the end item.- The end item shall be examined in accordance with the defects at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.3.1, 4.3.3.2 and 4.3.3.3. The lot size for purposes of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of yards for the examination in 4.3.3.1, 4.3.3.2 and 4.3.3.3.

4.3.3.1 Yard-by-yard examination.- The required yardage of each roll shall be inspected and visual defects classified as listed. The cloth shall be so alternated that on 1 of 3 rolls in the sample, the required yardage shall be examined on the inside and that on the other 2 of 3 rolls, the required yardage shall be examined on the outside. The defects listed below shall be counted regardless of their proximity to each other except where 2 or more defects represent a single local condition of the cloth, in which case, only the more serious defects shall be counted. A continuous defect shall be counted as 1 defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard. The inspection level shall be level I of MIL-STD-105. The acceptable quality level shall be 1.5 defects per 100 units. The number of rolls from which the sample is to be selected shall be in accordance with table III. Not more than 1 roll shall be selected from any container. An approximate equal number of yards shall be examined from each roll.

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Defects

Any cut, hole, tear or abrasion mark.
Any blister, lump or foreign matter (any size).
Any spot, stain or streak.
Uneven coating - thin area where coating is missing or insufficient.
Any overheavily coated area.
Any evidence of delamination.
Any crease or wrinkle - resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure.
Width, less than minimum specified.
Overall uncleanness.
Objectionable odor.

NOTE: Aromatic odors usually associated with specific types of coating compounds or solvents shall not be regarded as objectionable.

Not black on outside.
Tackiness (coating will adhere and not readily unroll).
Outside not marked as specified.
Edges rolled, folded, cut, wavy or distorted.

4.3.3.2 Examination for length and put-up.-

* 4.3.3.2.1 Individual rolls.- Each roll in the sample shall be examined for the defects listed below. The sample unit for this examination shall be 1 roll. The sample size and acceptance number shall be as shown in table III.

Defects

Any gross length of roll less than 1 yard of the gross length marked on the ticket.
Any gross length less than 80 or more than 100 yards long.
Any roll containing more than 3 splices.

TABLE III

Lot sizes in yards	Sample size in rolls	Acceptance number
1,300 or less 1/	3	0
1301 up to and including 3200	5	0

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TABLE III (cont'd)

Lot sizes in yards	Sample size in rolls	Acceptance number
3201 up to and including 8000	7	0
8001 up to and including 22,000	10	0
22,001 up to and including 110,000	15	1
110,001 and over	25	1

1/ If a lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.3.3.2.2 Combined lengths of sample units.- The lot shall be unacceptable if the total of the combined lengths of all rolls in the sample is less than the total of the combined lengths marked on the roll tickets. The rolls examined shall be those selected for examination of individual rolls in 4.3.3.2.1.

4.3.3.3 Examination of preparation for delivery.- An examination shall be made to determine that the fully packed shipping containers conforms to the packaging, packing and marking requirements of PPP-P-51. The sample unit shall be 1 shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closing defects. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

* 4.3.4 Testing of the end item.- Whenever applicable, the methods of testing specified in CCC-T-191 and those listed in table V herein shall be followed. The requirements specified in Section 3 apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test method. The lot size shall be expressed in units of 1 linear yard. The sample unit for test purposes shall be 1-1/2 yards, full width of finished fabric. The sample size shall be in accordance with table IV. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. For those characteristics wherein test results are reported as pass or fail, there shall be no failure of any test specimen to meet the requirements as specified.

TABLE IV

Lot size (yards)	Sample size
800 or less	2
8001 to 22,000	3
22,001 and over	5

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* TABLE V - Instructions for testing

Characteristic	Requirement	Test method	Number of determinations per sample unit	Results reported as Pass or fail	Results reported as Numerically to nearest
Coated fabric:					
Total weight	3.3	5041	5		0.1 ounce
Breaking strength:					
Initial					
Warp (length)	Table II	5102	5		1.0 pound
Filling (width)	Table II	5102	5		1.0 pound
After accelerated weathering					
Warp (length)	Table II	5102 and 4.4.3	5		1.0%
Filling (width)	Table II	5102 and 4.4.3	5		1.0%
Ply adhesion					
Warp (length)	Table II	4.4.2	1 per layer		0.5 pound
Filling (width)	Table II	4.4.2	1 per layer		0.5 pound
Permeability, liters per sq meter per 24 hours	Table II	5640	3		liter
Accelerated weathering	Table II	4.4.3	1	X	

4.4 Test methods.-

4.4.1 Test conditions.- Results of physical tests obtained under testing conditions defined in FED. TEST METHOD STD. NO. 406, FED. TEST METHOD STD. NO. 601 or CCC-T-191 will be acceptable except in case of dispute; in dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in CCC-T-191.

4.4.2 Ply adhesion.- The ply adhesion shall be determined as specified in method 8011 of FED. TEST METHOD STD. NO. 601. The test shall be applied in both the warp (length) and filling (width) directions of the coated, laminated cloth. One adhesion test shall be made for each of the six adhesion surfaces.

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4.4.3 Accelerated weathering.- The accelerated weathering test shall be performed in accordance with method 5804 of CCC-T-191 for a period of 200 hours. The specimens for this test shall be 1 by 6 inches. For the breaking strength test, the specimens shall be allowed to dry after the accelerated weathering test and then be conditioned at standard conditions for 4 hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The cloth, put-up as specified, shall be packaged in accordance with the applicable requirements for class 5 fabrics of PPP-P-51, except that in the put-up, the fabric shall not be secured from unwinding. Rolling of the cloth shall be done with the minimum tension practicable to produce a "soft roll".

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-51. Not more than two rolls shall be packed per container, with the rolls positioned two in width.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-51.

6. NOTES

6.1 Intended use.- The laminated coated fabric covered by this specification is intended to be used in the fabrication of decoy targets and as an air retaining repair material for pneumatic targets. It is a component of SC 1080-93-CL-E04, Repair Equipment, Pneumatic Target.

* 6.2 Ordering data.- Purchasers should exercise any desired options offered herein. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Width and length of coated fabric if other than specified (see 3.5).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. Whether preproduction sample is required (see 3.1).

* 6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices or alternatives, as indicated in PPP-P-51, may be exercised in the preparation for delivery of cloth.

* 6.3 Marginal notations.- The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made.

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Army - MO, WC

Project No. 8305-A308

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